QA Dashboard

AI Literacy for the Modern Workforce

Quality assurance on this project adapted Hardman's AI Evals framework to an L&D context. When the practitioner's core job shifts from authoring content to evaluating AI-generated content, the critical skills become specification (defining "good" precisely enough for AI to produce it) and verification (reliably detecting when output fails that bar). The method is a binary pass/fail cycle: select an output type, collect samples, define checks, score, diagnose root cause (specification failure vs. generalization failure), revise the specification, re-score, and repeat on a schedule.

Applied across four output categories and six data files, it ran as ten structured verification passes, a triple-validation protocol for the data files, and an eight-check capability framework, all recorded below. Everything the passes surfaced was resolved before or during the build.

TEN VERIFICATION PASSES

QA PASS	SCOPE · SAMPLE	RESULT
Capability check logs (M1–M4) Inline pass/fail checks during authoring	Narrative, scenario stems, feedback, KC items 4 modules	All passed; outstanding items resolved before build handoff.
Knowledge-check evaluation Binary checks: scenario-based, application-level, plausible distractors, consequence feedback, no cueing	16 KC items across 4 modules 16 items	KC-1.1–1.3 revised for distractor balance; KC-2.3/2.4 for forward-reference; KC-4.4 Option C for length parity.
Practice-activity traceability audit Each P-activity vs. action-map and content D-codes	12 practice activities (P1–P12) 12 items	All 12 match across both sources in behavior targets and descriptions.
KC–objective traceability Each KC stem vs. its referenced performance objective	16 KCs against 17 objectives 16 items	All 16 correctly assess their referenced objectives.
Data-file triple validation Schema, value verification vs. sources, cross-file consistency	6 JSON files 6 files × 3 passes	All validated; $55 → $54 and citation page-range corrections applied.
Cross-cutting integration check (1) Event model, state shape, completion logic, consumer mapping	5 component specs + 6 data files 28 edits across 6 files	95 event keys cataloged; 37+ state keys reconciled; master event-key table produced.
Cross-cutting integration check (2) Content flow, terminology, tone, D-code trace, color batch fix	4 content docs + 2 planning + 4 specs 6 tasks	All clean; six 03a alignment-matrix discrepancies corrected; two stale markers flagged.
JSX artifact audit Reconcile interactive diagrams vs. the built codebase	2 JSX files against src/ 18 discrepancies	5 major, 13 minor found and resolved; downstream doc fixes applied.
Citation verification pass Every inline statistic vs. reference-quotes.md and source papers	21 statistics across 4 modules 21 items	One known discrepancy (81% vs 84% median savings) confirmed resolved per the Figure 6 methodology.
Reference item (R1–R7) design review PDF assets vs. design system; content accuracy and register	7 reference items at 10 placements 7 items	All 7 redesigned in the project design system; contextual placement verified.

TRIPLE-VALIDATION DATA PROTOCOL

Six JSON data files each ran three dedicated validation passes, a stricter standard than a single-pass eval, because learners draw conclusions directly from the dashboards this data powers.

1 Schema validation

Field names, data types, required vs. optional fields, and structural consistency across array entries.

2 Value verification

Every numeric value and quoted string traced to a specific page, table, or figure in the source paper, cross-checked against reference-quotes.md.

3 Cross-file consistency

Values shared across files (e.g., the 81% median-savings figure) verified for consistency; each file's consuming components confirmed against the specs.

DATA FILE	SOURCE	KEY CONTENT	STATUS
augmentation-automation.json	Handa et al.	Occupation/task breakdowns; 57%/43% augmentation-automation split	● Pass
productivity-gains.json	Tamkin & McCrory	Task time reductions; 81% median savings (Fig. 6 caption, p. 13)	● Pass · 2 corrections
geographic-adoption.json	Appel, McCrory & Tamkin	Country-level usage rates; institutional-inertia indicators	● Pass
wef-skill-demand.json	WEF Future of Jobs 2025	Fastest-growing/declining skills; employer transformation expectations	● Pass
interviewer-findings.json	Anthropic Interviewer	86% time savings; 65% satisfaction; 69% stigma; 48% career transition	● Pass
adoption-trends.json	Multiple sources	Longitudinal adoption rates; interaction-mode temporal data	● Pass

EIGHT-CHECK CAPABILITY FRAMEWORK

Each check is Capability (does the output do what it should?) or Risk (what can go wrong that the learner cannot detect?). Risk checks matter most in learning contexts, because learners consume content without an answer key.

Scenario authenticity CAPABILITY

Does the stem present a recognizable workplace situation with a job title, context, and decision point?

Application-level cognitive demand CAPABILITY

Does the item test applying a concept to a novel situation rather than recalling a definition?

Distractor plausibility RISK

Are all non-preferred options actions a reasonable professional might consider? (No trick or absurd choices.)

Correct-answer cueing absence RISK

Is the preferred option free of length, specificity, or hedging cues that distinguish it without content knowledge?

Consequence-based feedback CAPABILITY

Source fidelity RISK

Does each option branch lead to a realistic observed outcome rather than a "Correct" / "Try again" label?

Does every cited statistic match its source paper (checked against reference-quotes.md)?

Register consistency CAPABILITY

Does the text hold the "knowledgeable peer" second-person register without slipping into lecture or compliance tone?

Forward-reference avoidance RISK

Does the item avoid requiring knowledge from a module the learner has not yet completed?

The single-evaluator limit

The structured passes above were run by the same person who authored the content, so specification failures are hardest to detect where the specifier and the evaluator share a mental model. The same conflict was named in the capstone pilot, where the instructor was also the researcher (Ritchot, 2025, p. 39). Its controls were success criteria fixed in advance, critical feedback kept in the analysis, anonymous labeling of responses, and triangulation across four data sources (pp. 39–40). The pass/fail checks below carry the first two forward. The independent model review stands in for the third. An independent review of both the code and the content by a different model (Claude Fable) provided a second pass that did not share that mental model and surfaced issues that were then fixed. That does not replace what an organizational deployment should add: the evaluation role separated from authoring, with inter-rater reliability established for the check framework. But it is more than a single set of eyes, and the binary pass/fail format is designed to transfer: it is specific enough that a different evaluator would reach substantially similar conclusions. That transferability is the practical test of whether the methodology scales beyond one person.

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